



05506026

Digital Image Processing

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MORPHOLOGICAL IMAGE PROCESSING



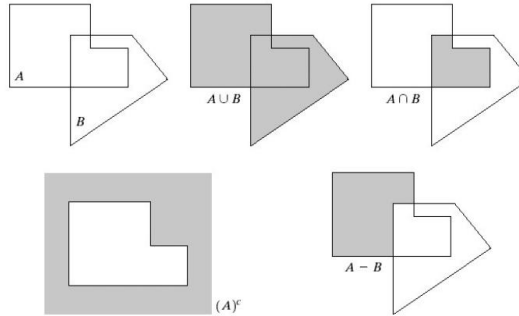
MORPHOLOGICAL IMAGE PROCESSING

Morphology

- ▶ Morphology is a word that is commonly used to describe form and structure of animals and plants
- ▶ In digital image processing, morphology is used in the context of mathematics as a tool for extracting image components
- ▶ Morphology is useful in the representation and description of region shape such as boundaries, skeletons, and convex hull

Morphology

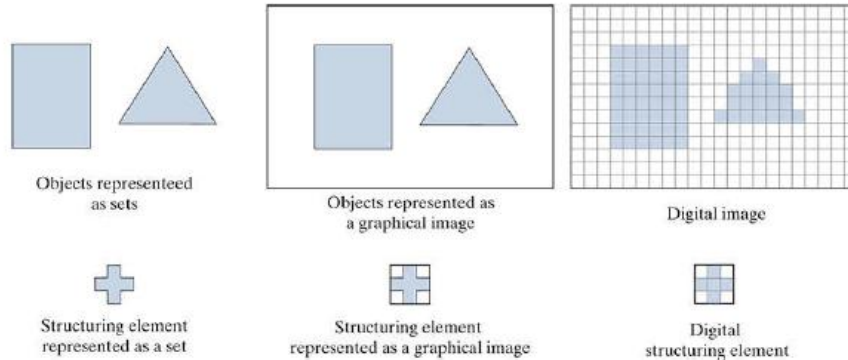
- Morphology involves Set Theory



- There are two types of sets of pixels: Objects and Structuring Elements (SE)

Morphology

- ▶ In a binary image, Objects are sets of foreground pixels
- ▶ Structuring Elements can be both foreground and background pixels

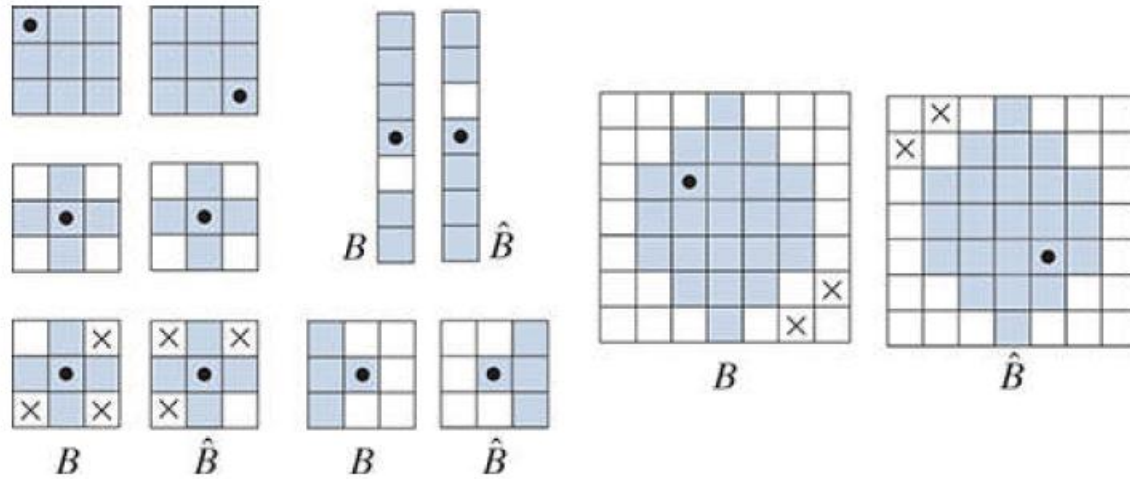


Morphology

- ▶ Structuring Elements has some annotations as follow
 - ▶ • denotes the origin of the set
 - ▶ X denotes 'don't care' elements
- ▶ The reflection of an SE B about its origin is $\hat{B}$ defined as

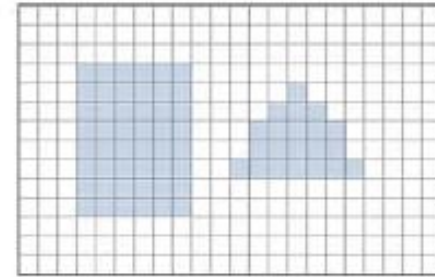
$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$

Morphology



Morphology Operations

- ▶ Morphology operations are operations where the set of SE is slid through the object, similar to spatial filtering operations
- ▶ If the conditions are met, the operations will be executed on the object according to the origin of SE



Digital image



Digital structuring element



Erosion and Dilation

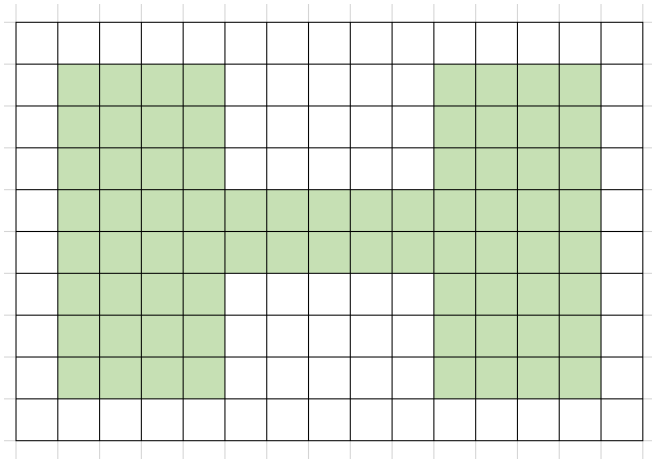
Erosion

- ▶ If A is a set of foreground pixels (object) and B is structuring elements, the erosion of A by B is defined as

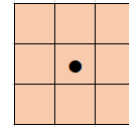
$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

- ▶ Where Z are foreground values (1)
- ▶ Meaning that, the condition of erosion is all foreground pixels in B is a subset of A
- ▶ The operation of erosion is placing a foreground pixel value to the pixel which equivalent to the origin of SE

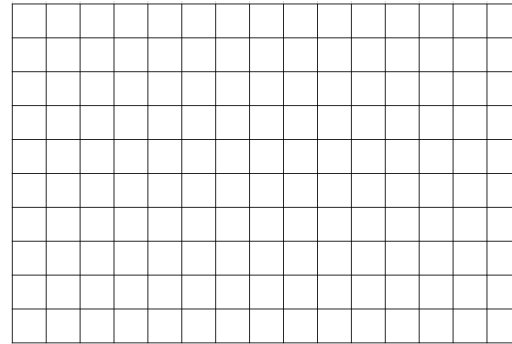
Erosion



Object

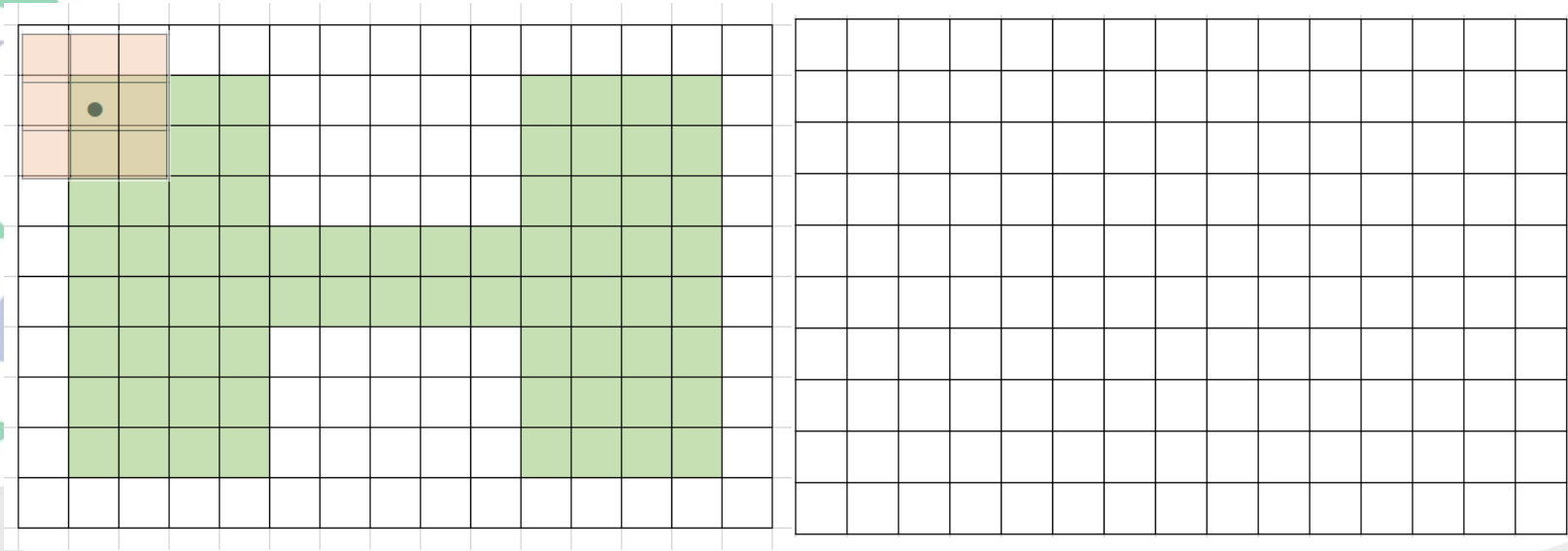


Structuring Elements

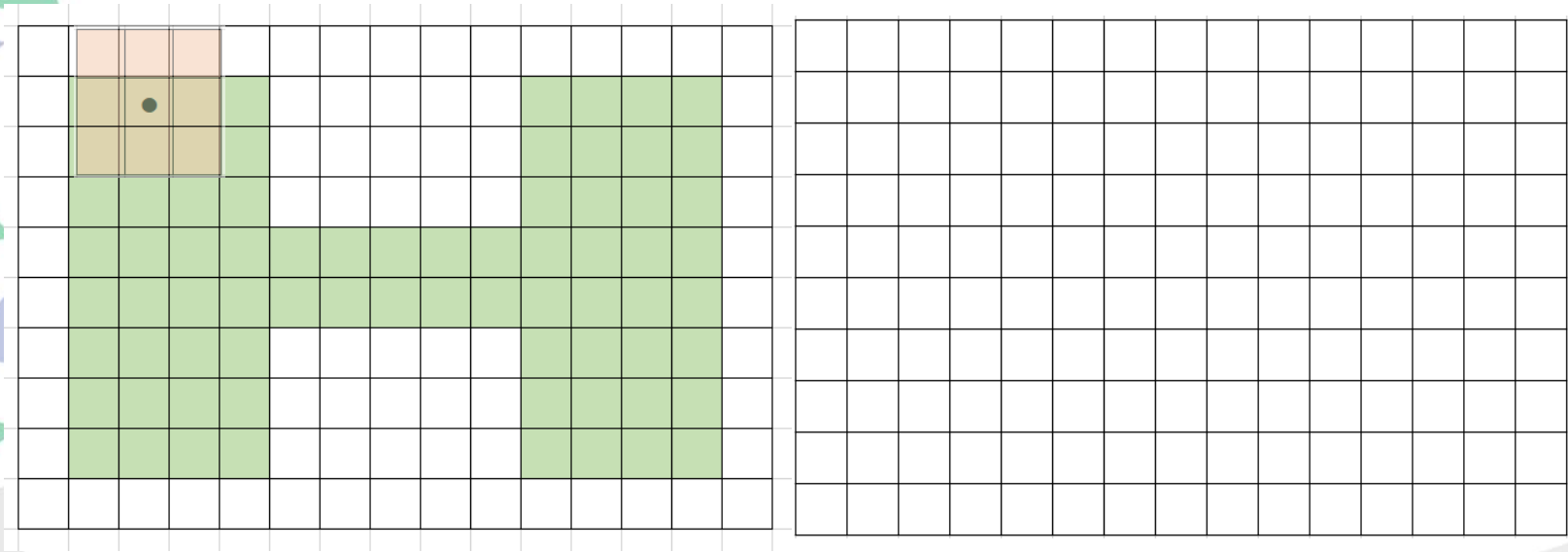


New image

Erosion

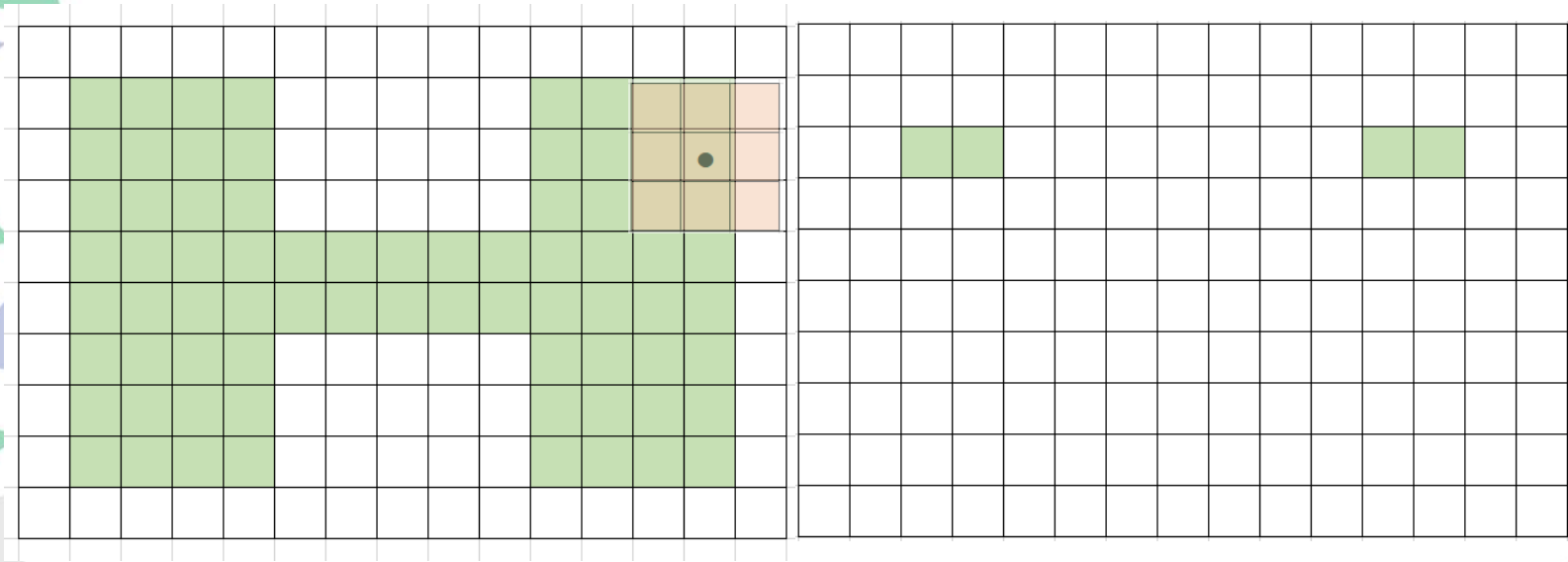


Erosion

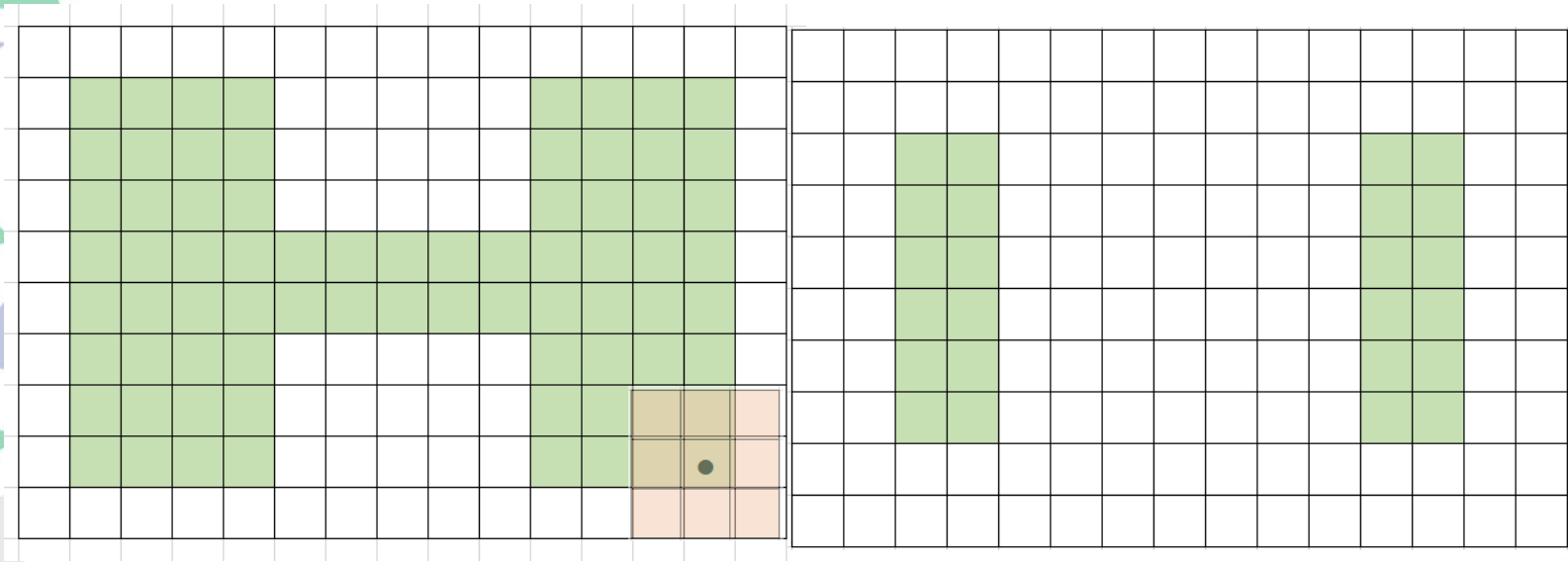


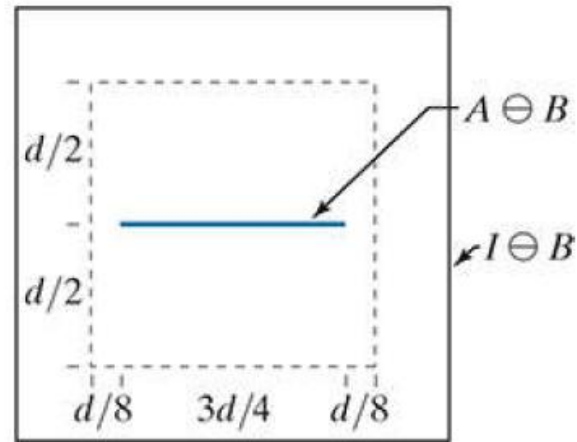
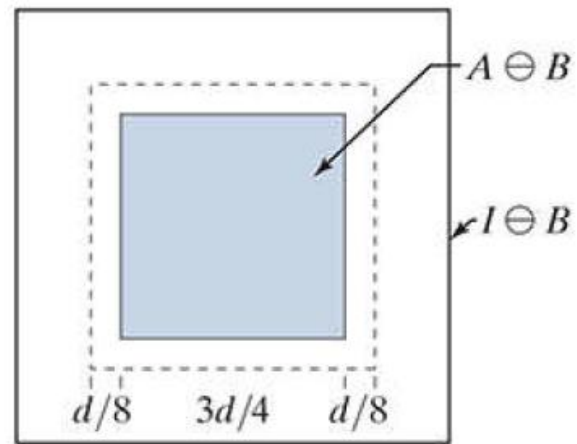
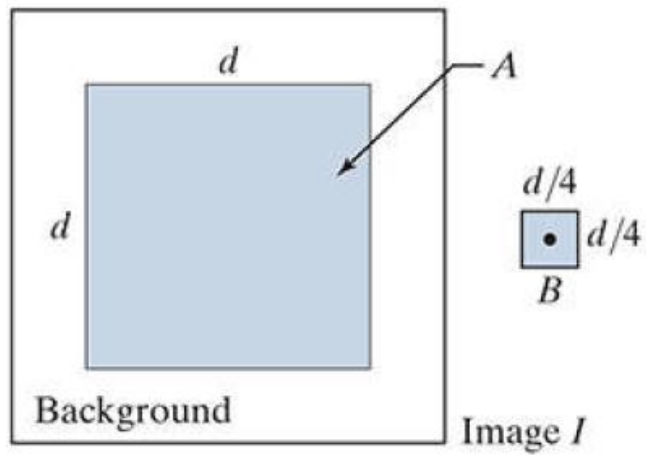
A 10x10 grid. A green shape is composed of 24 cells, starting from row 2, column 2 and extending to row 8, column 8. An orange shape is composed of 6 cells, starting from row 8, column 8 and extending to row 10, column 10. A black dot is located at row 9, column 9.

Erosion

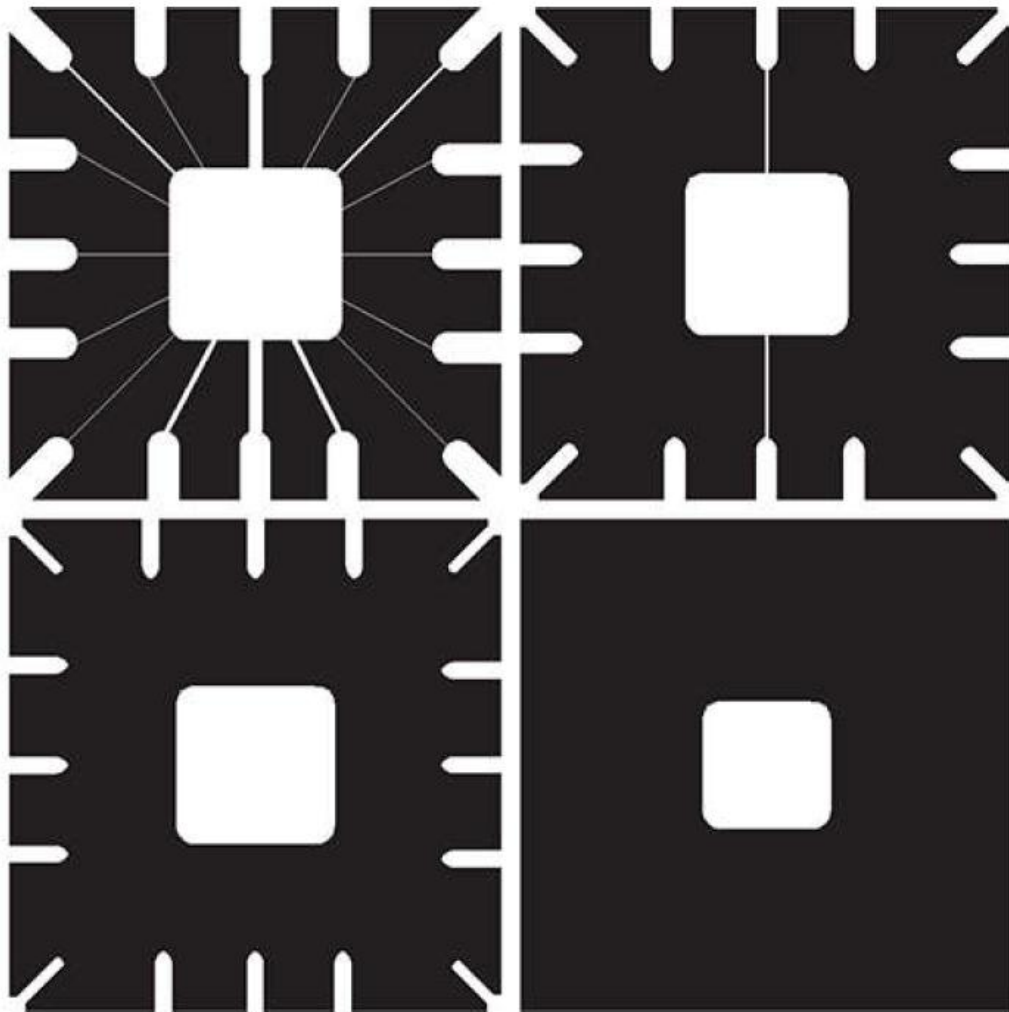


Erosion





Object



**Eroded with
11x11 square
SE**

**Eroded with
15x15 square
SE**

**Eroded with
45x45 square
SE**

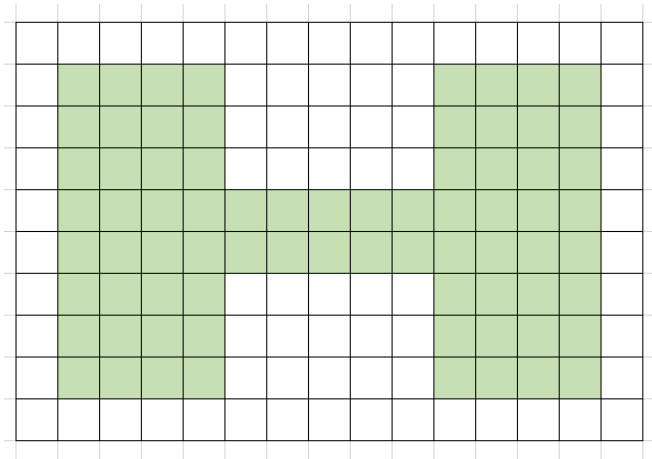
Dilation

- ▶ If A is a set of foreground pixels (object) and B is structuring elements, the erosion of A by B is defined as

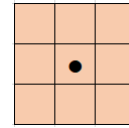
$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$$

- ▶ Where Z are foreground values (1)
- ▶ Meaning that, the condition of erosion is at least foreground pixels in $\hat{B}$ is a subset of A
- ▶ The operation of erosion is placing foreground pixel values to every pixels which equivalent to the SE

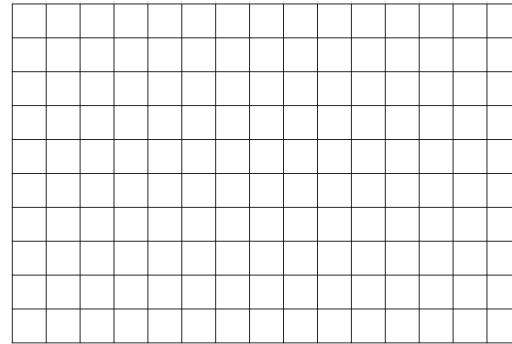
Dilation



Object

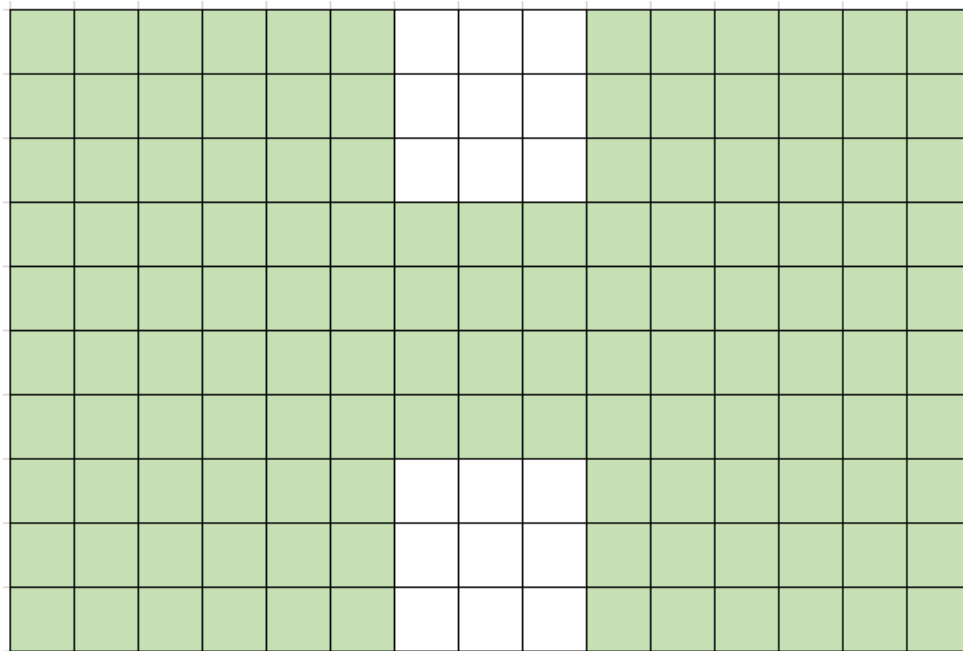


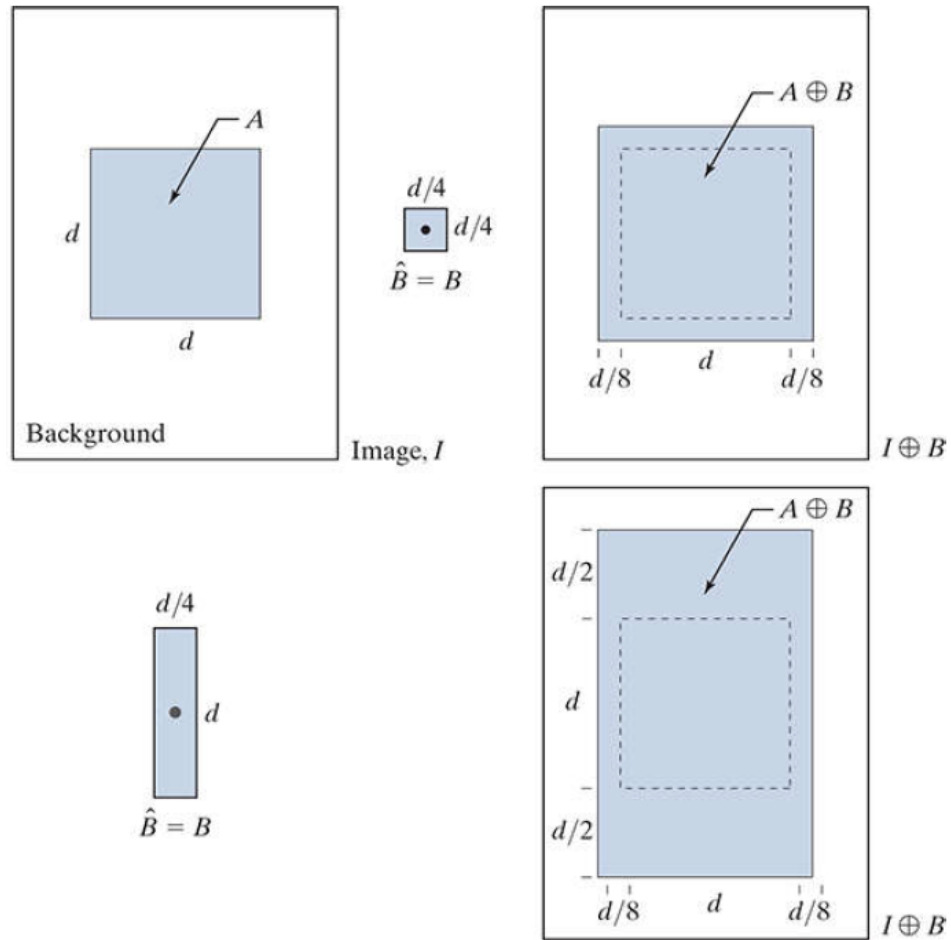
Structuring Elements



New image

Dilation





Dilation

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



1	1	1
1	1	1
1	1	1

Duality

- ▶ Erosion and Dilation are duals of each other

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

And

$$(A \oplus B)^c = A^c \ominus \hat{B}$$



Opening and Closing

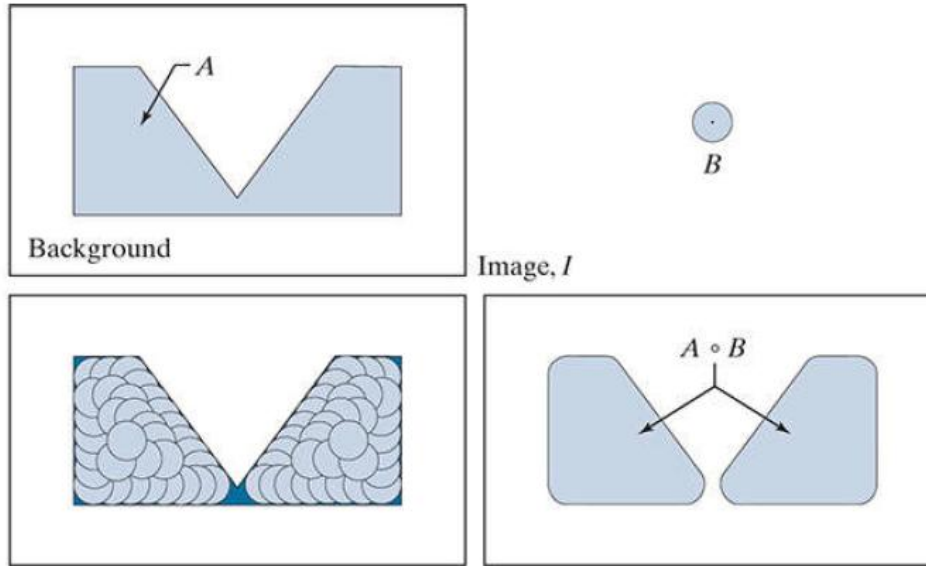
Opening

- ▶ The opening of set A by structuring element B is defined as

$$A \circ B = (A \ominus B) \oplus B$$

- ▶ Thus, the opening A by B is the erosion of A by B, followed by a dilation of the result by B.

Opening



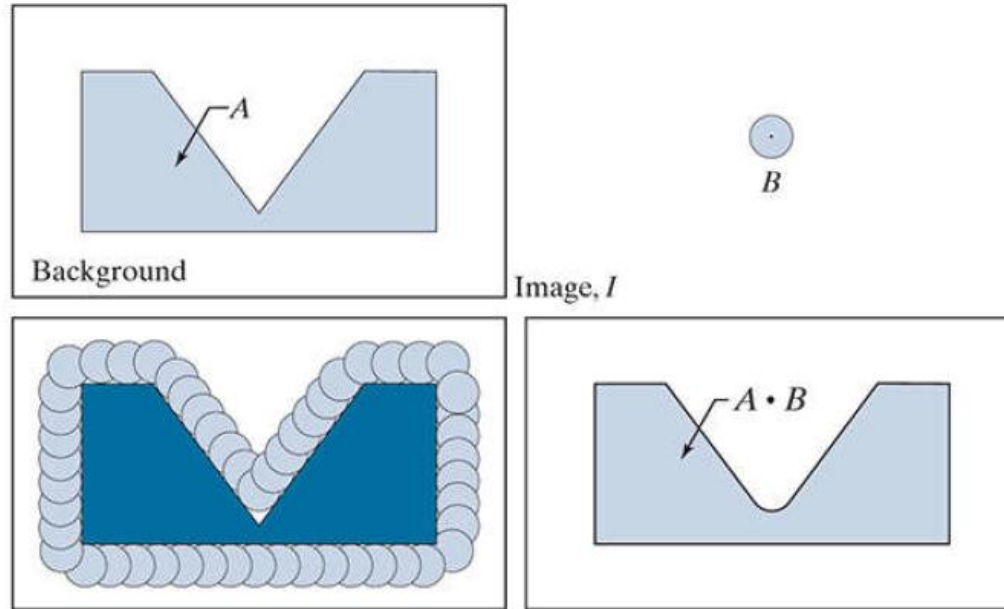
Closing

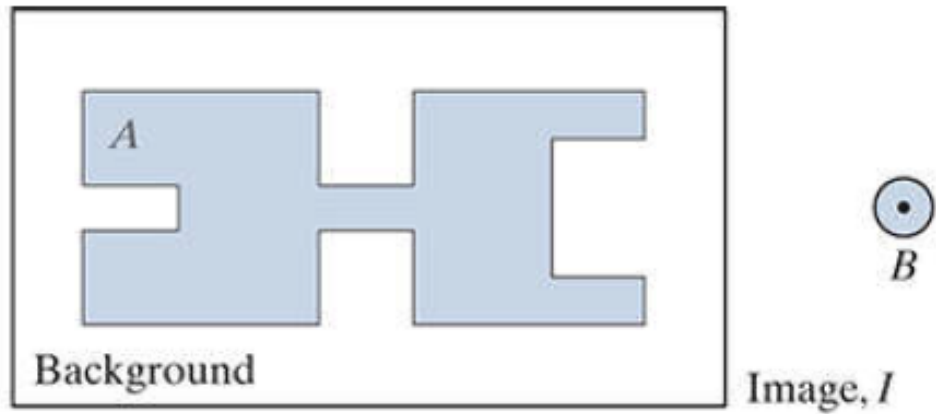
- ▶ The closing of set A by structuring element B is defined as

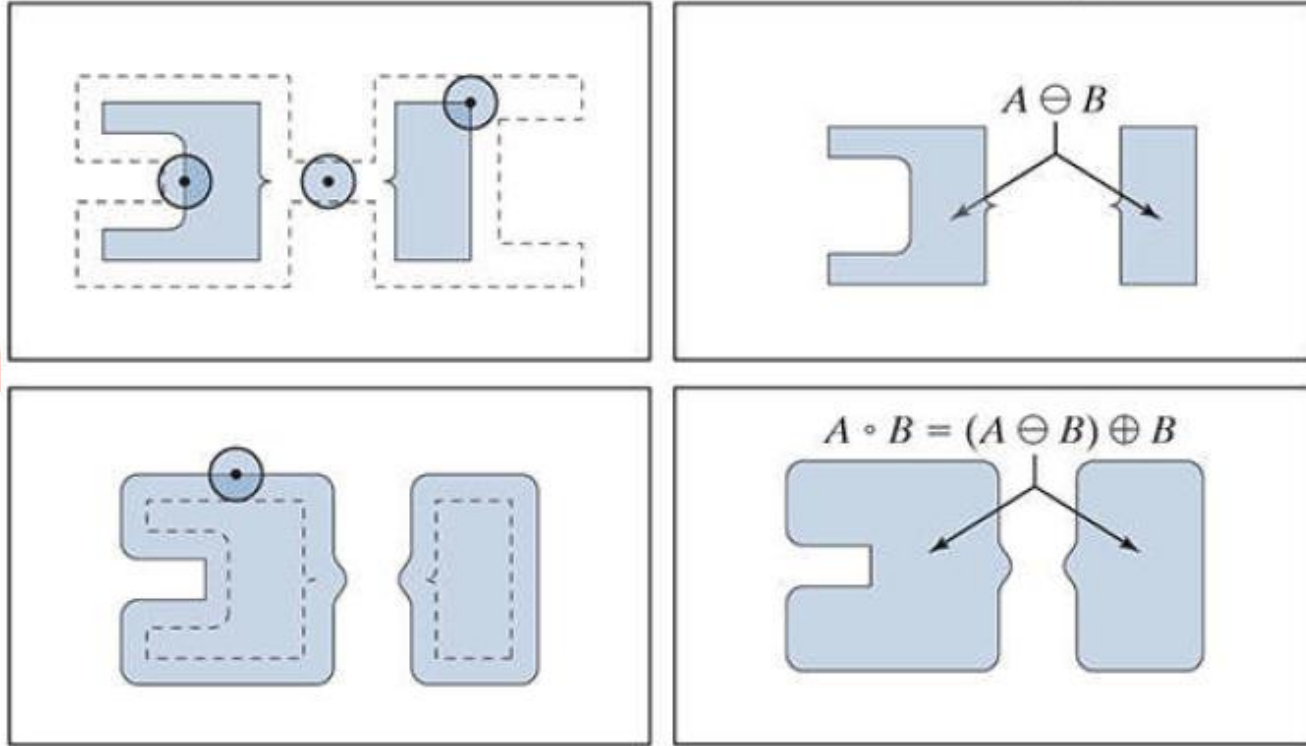
$$A \bullet B = (A \oplus B) \ominus B$$

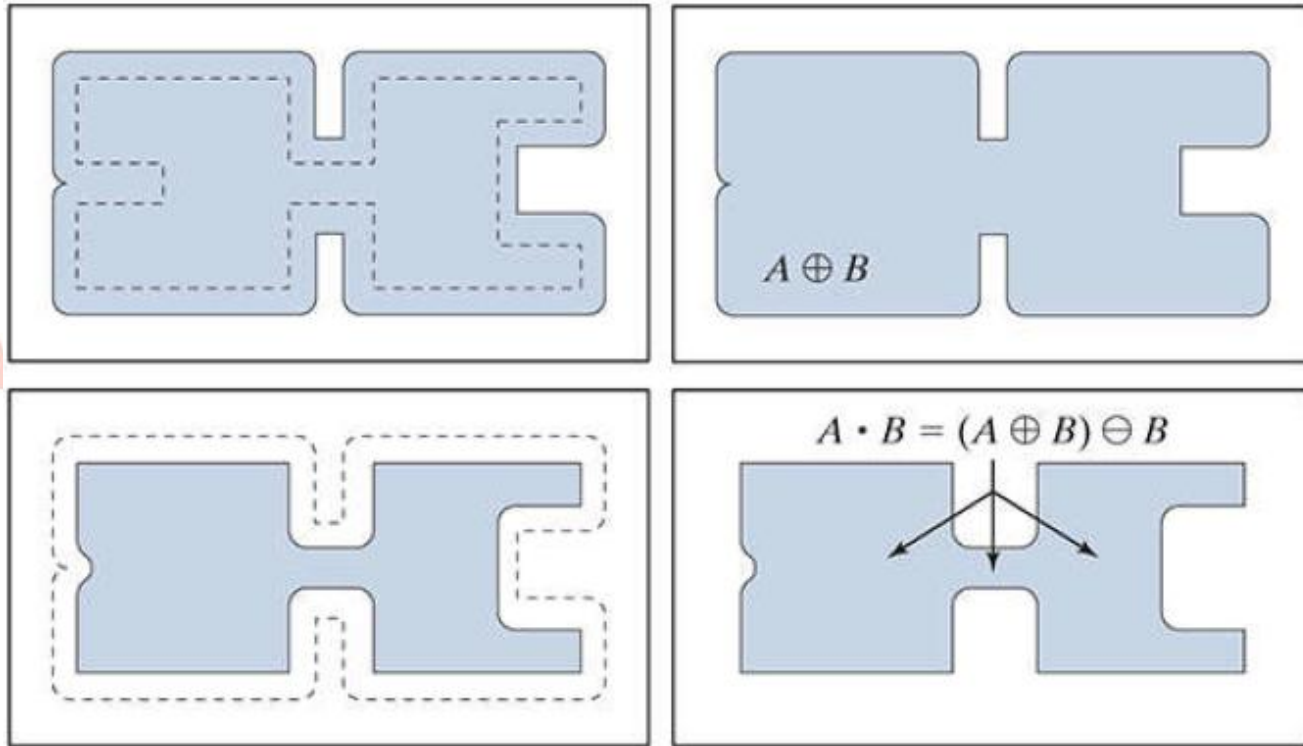
- ▶ Thus, the closing of A by B is the dilation of A by B, followed by erosion of the result by B.

Closing











A (foreground pixels)

1	1	1
1	1	1
1	1	1

B



A (foreground pixels)

1	1	1
1	1	1
1	1	1

B

$$A \ominus B$$





$$A \ominus B$$

1	1	1
1	1	1
1	1	1

 B

$$(A \ominus B) \oplus B = A \circ B$$





$$(A \ominus B) \oplus B = A \circ B$$

1	1	1
1	1	1
1	1	1

 B

$$(A \circ B) \oplus B$$





$$(A \circ B) \oplus B$$

$$B$$

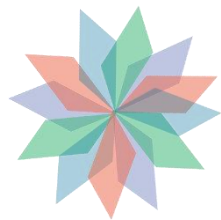
1	1	1
1	1	1
1	1	1

$$[(A \circ B) \oplus B] \ominus B = (A \circ B) \cdot B$$





Hit-Or-Miss Transform

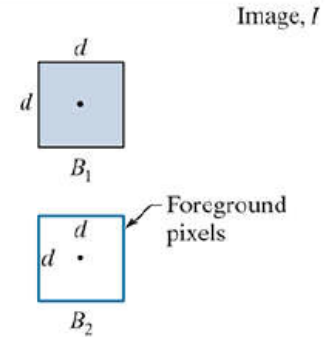
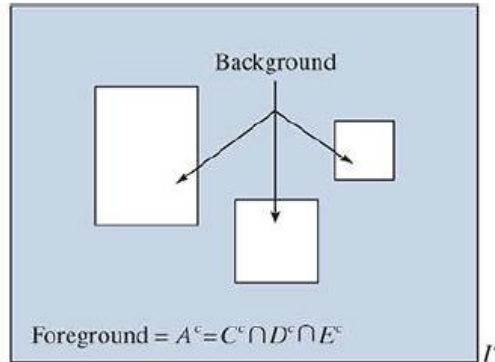
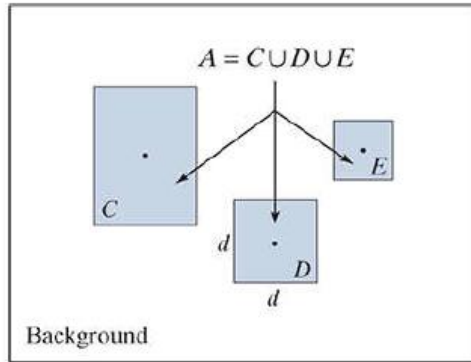


Hit-Or-Miss Transform

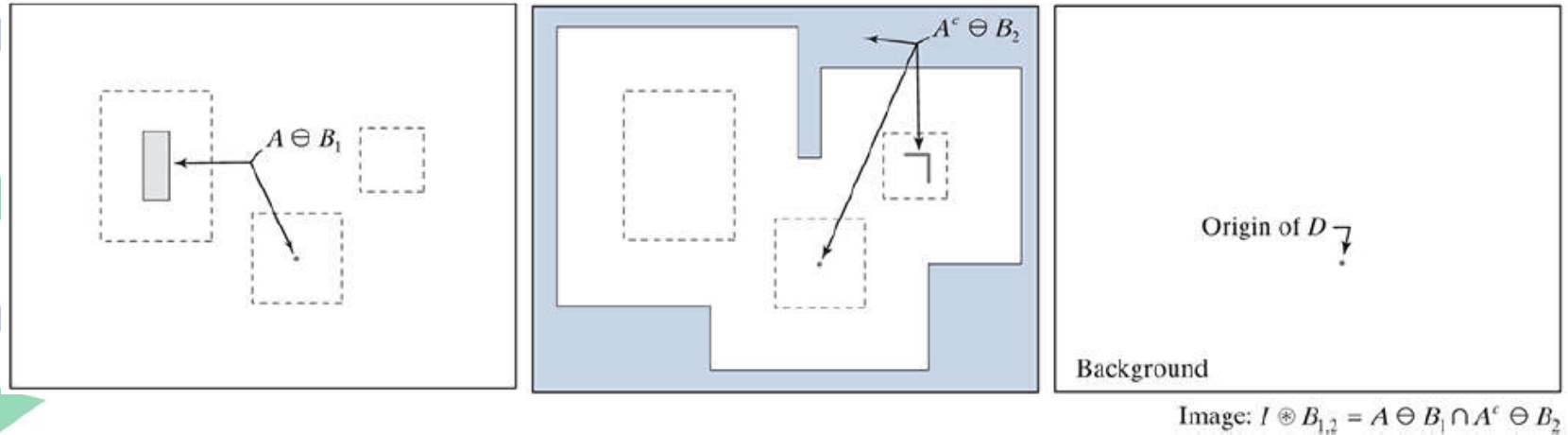
- ▶ Hit-or-miss transform (HMT) is a basic tool for shape detection
- ▶ Let A is a set of foreground pixels in a binary image I , HMT utilises TWO structuring elements: -
 - ▶ B_1 for detecting shapes in foreground
 - ▶ B_2 for detecting shapes in background
- ▶ The HMT of image I is defined as

$$\begin{aligned} I \circledast B_{1,2} &= \{z \mid (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c\} \\ &= (A \ominus B_1) \cap (A^c \ominus B_2) \end{aligned}$$

Hit-Or-Miss Transform

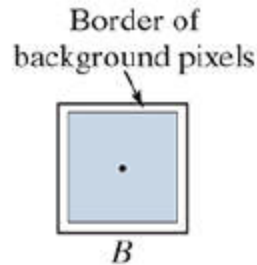


Hit-Or-Miss Transform



Hit-Or-Miss Transform

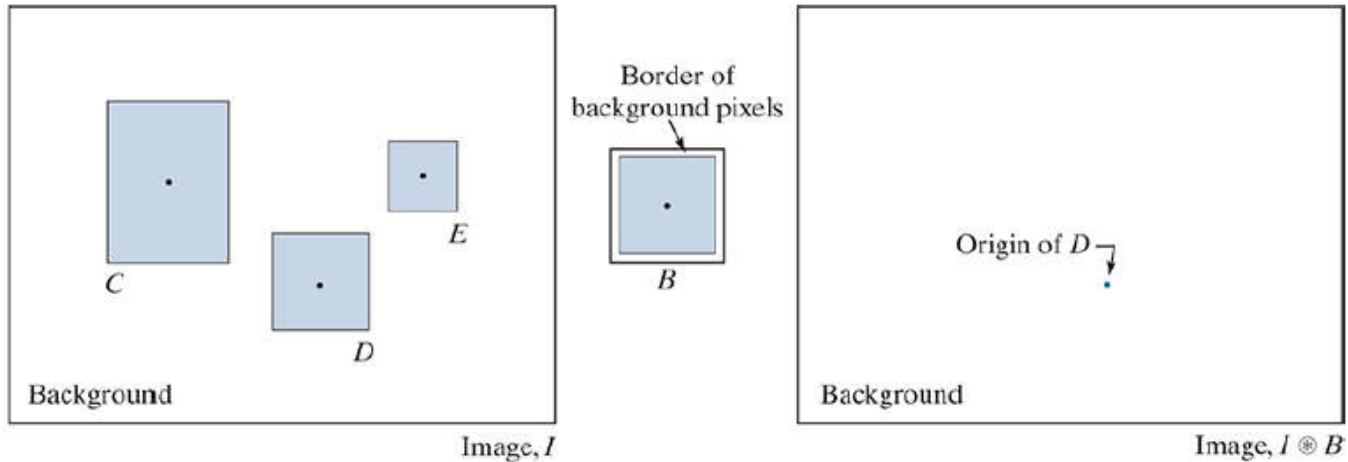
- ▶ HMT can be done by only single SE by combining B_1 and B_2 to be a SE with one-pixel border

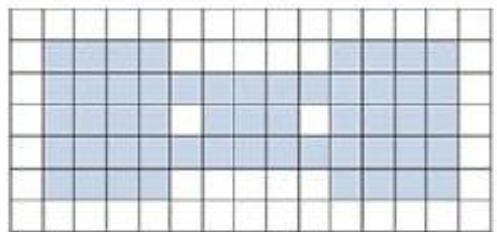


- ▶ HMT can now be restated into an erosion as

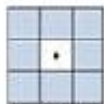
$$I \circledast B = \{z \mid (B)_z \subseteq I\}$$

Hit-Or-Miss Transform

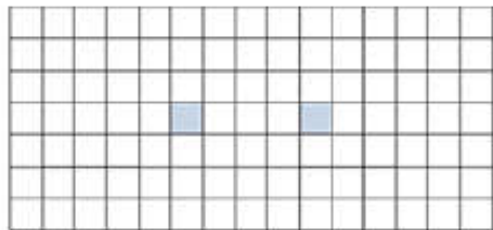




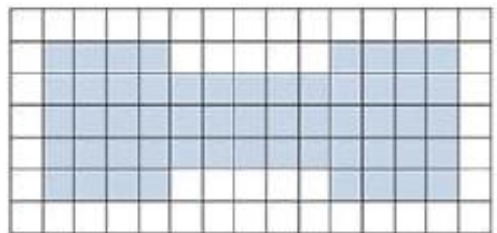
Image, I



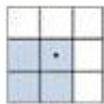
B



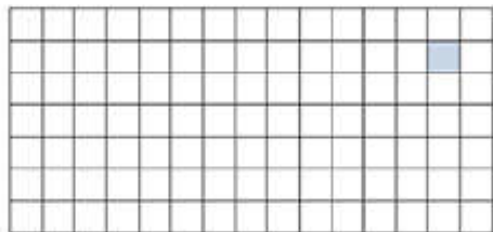
Image, $I * B$



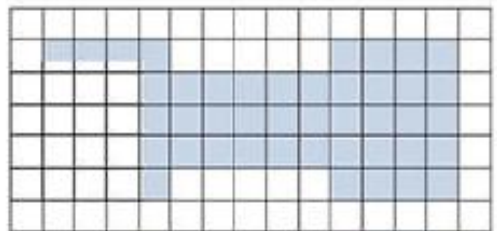
Image, I



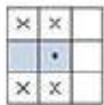
B



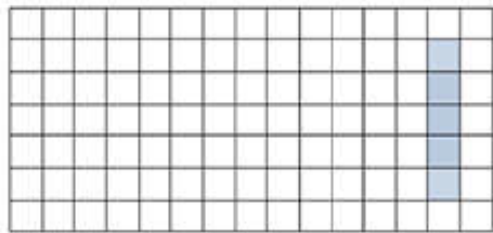
Image, $I * B$



Image, I



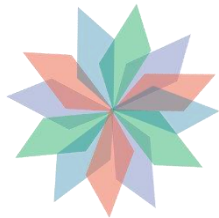
B



Image, $I * B$

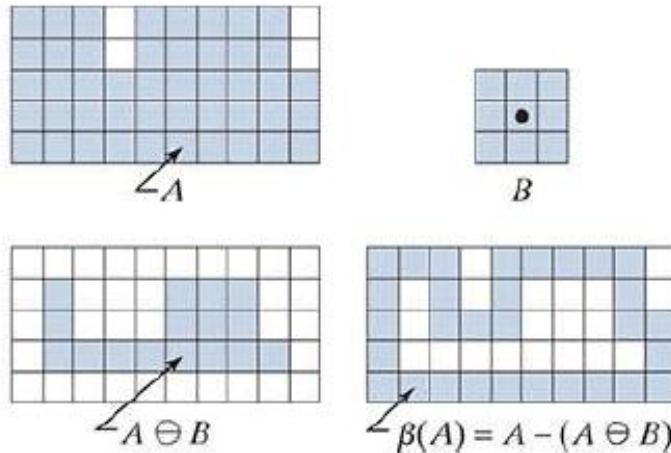


Some Other Basic Morphological Algorithms

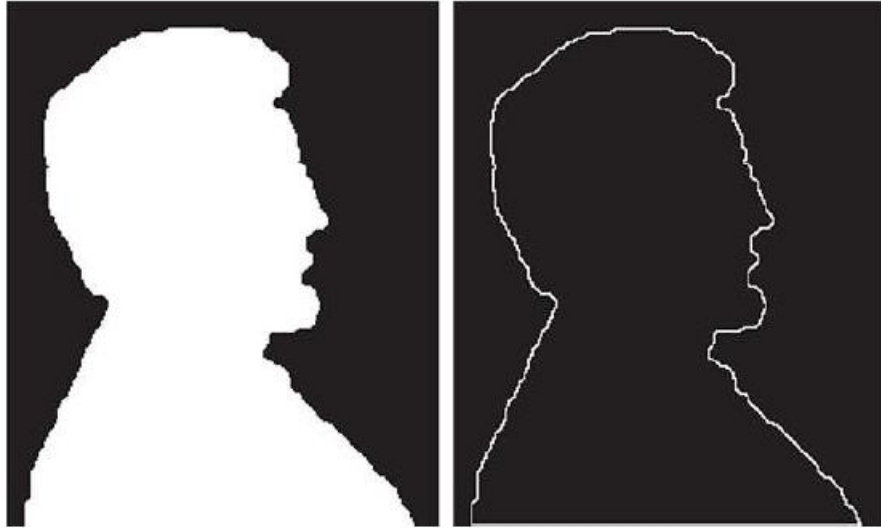


Boundary Extraction

$$\beta(A) = A - (A \ominus B)$$



Boundary Extraction

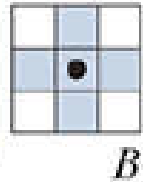


Hole Filling

- ▶ Given X is a point in a hole, we can fill this hole by repeating the step

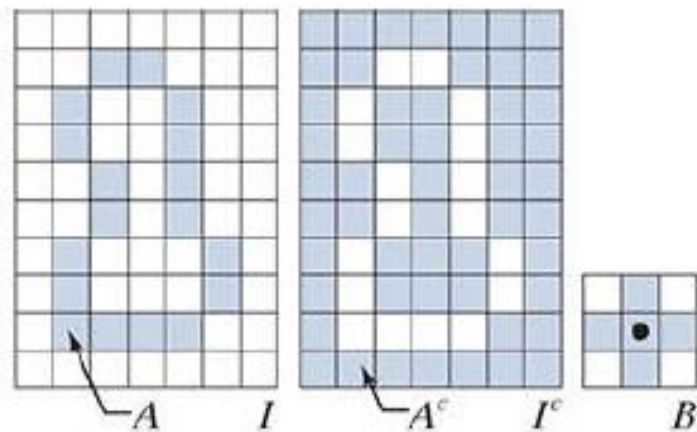
$$X_k = (X_{k-1} \oplus B) \cap I^c \quad k = 1, 2, 3, \dots$$

- ▶ With SE

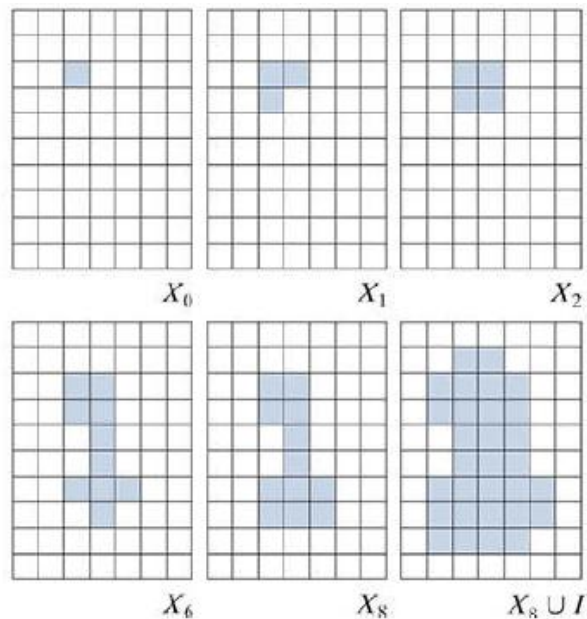


- ▶ Until $X_k = X_{k-1}$

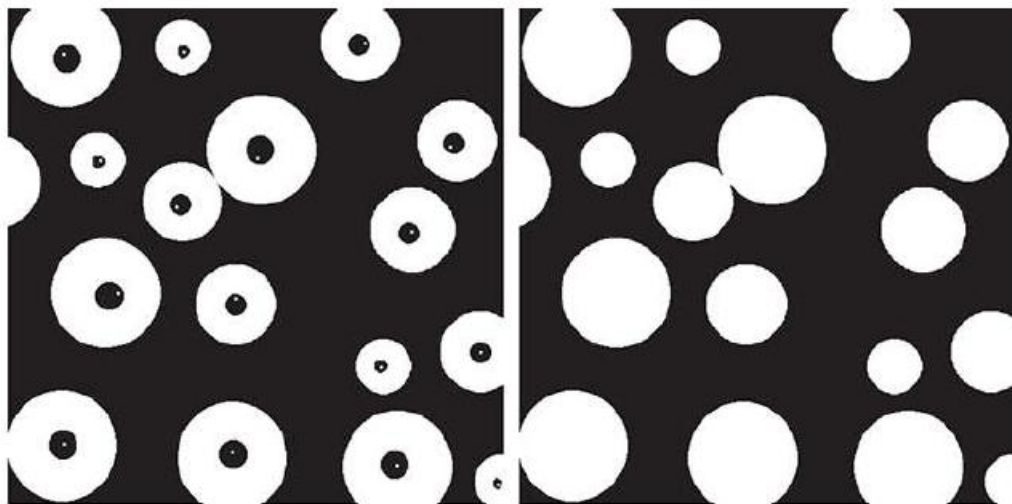
Hole Filling



Hole Filling



Hole Filling

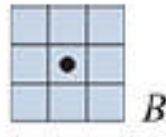


Extraction of Connected Components

- ▶ Given X is a point in an object, we can extract the connected components by repeating the step

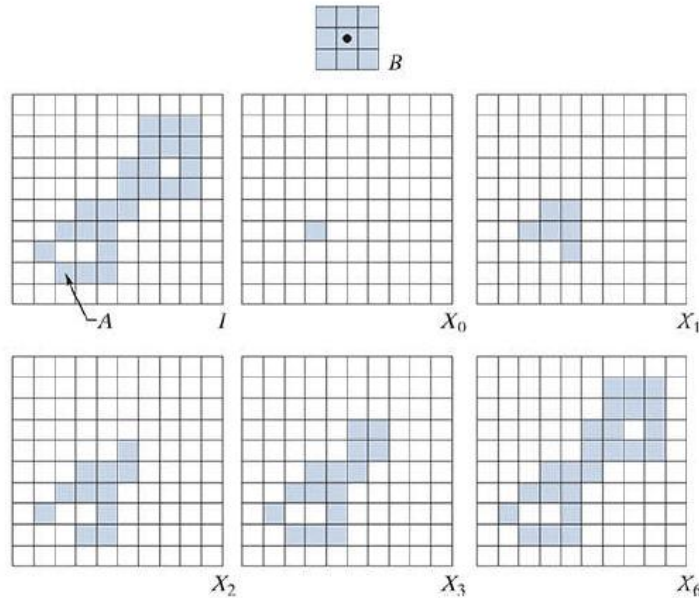
$$X_k = (X_{k-1} \oplus B) \cap I \quad k = 1, 2, 3, \dots$$

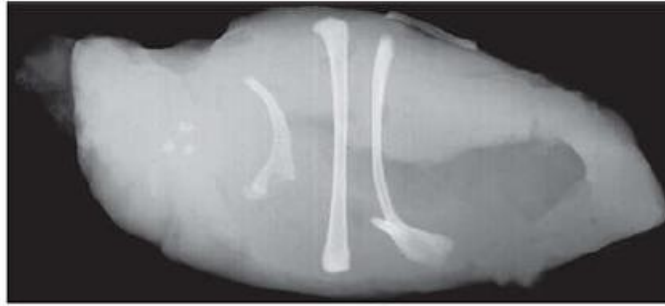
- ▶ With SE



- ▶ Until $X_k = X_{k-1}$

Extraction of Connected Components

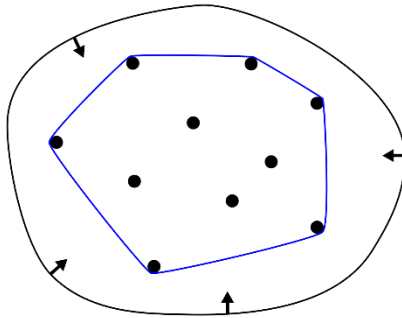




Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Convex Hull

- ▶ A set, S , of points in the Euclidean plane is said to be convex if and only if a straight-line segment joining any two points in S lies entirely within S
- ▶ The convex hull, H , of S is the smallest convex set containing S .

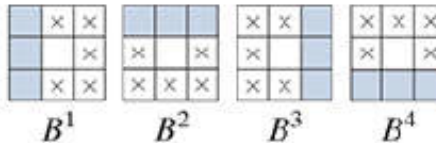


Convex Hull

- ▶ The convex hull of an object A in a binary image I can be extracted by repeating step

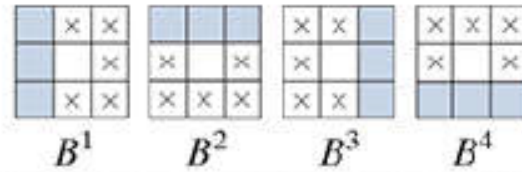
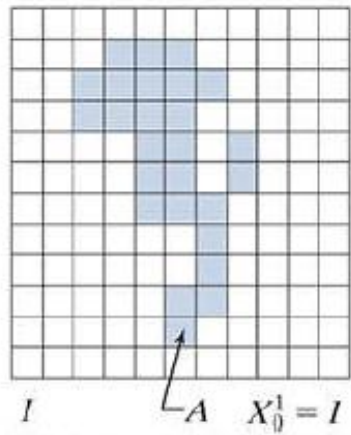
$$X_k^i = (X_{k-1}^i \otimes B^i) \cup X_{k-1}^i \quad i = 1, 2, 3, 4 \quad \text{and} \quad k = 1, 2, 3, \dots$$

- ▶ with 4 SEs B^i , $i = 1, 2, 3, 4$ as follow

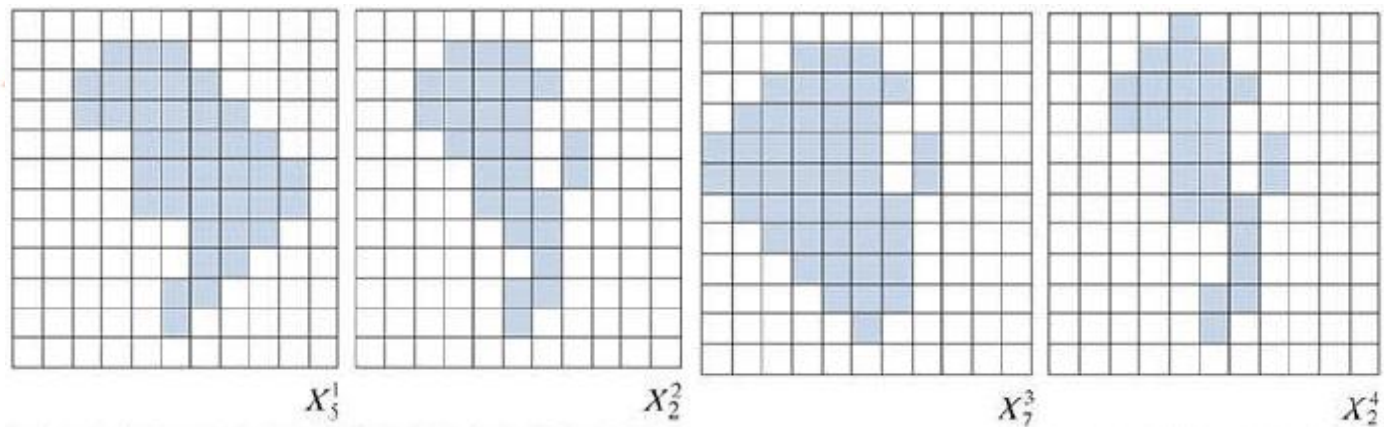


- ▶ Until $X_k^i = X_{k-1}^i$

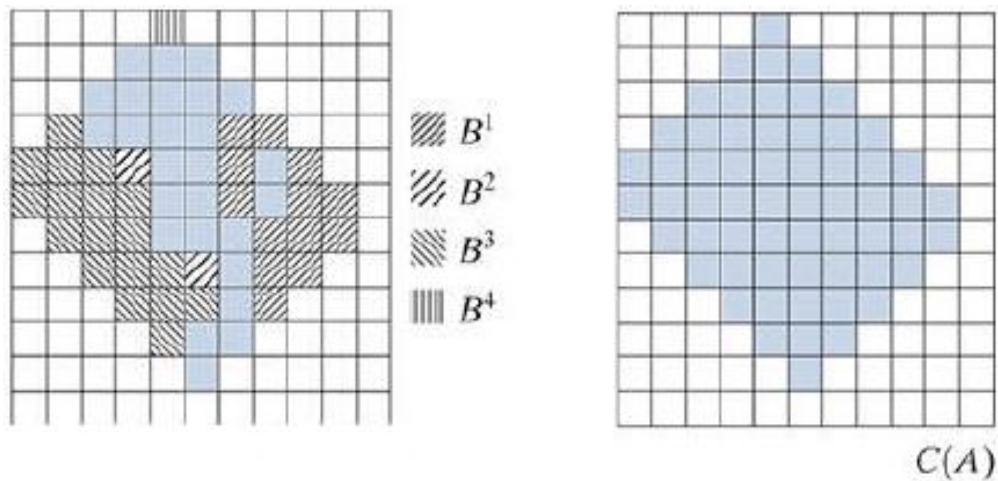
Convex Hull



Convex Hull



Convex Hull



Convex Hull

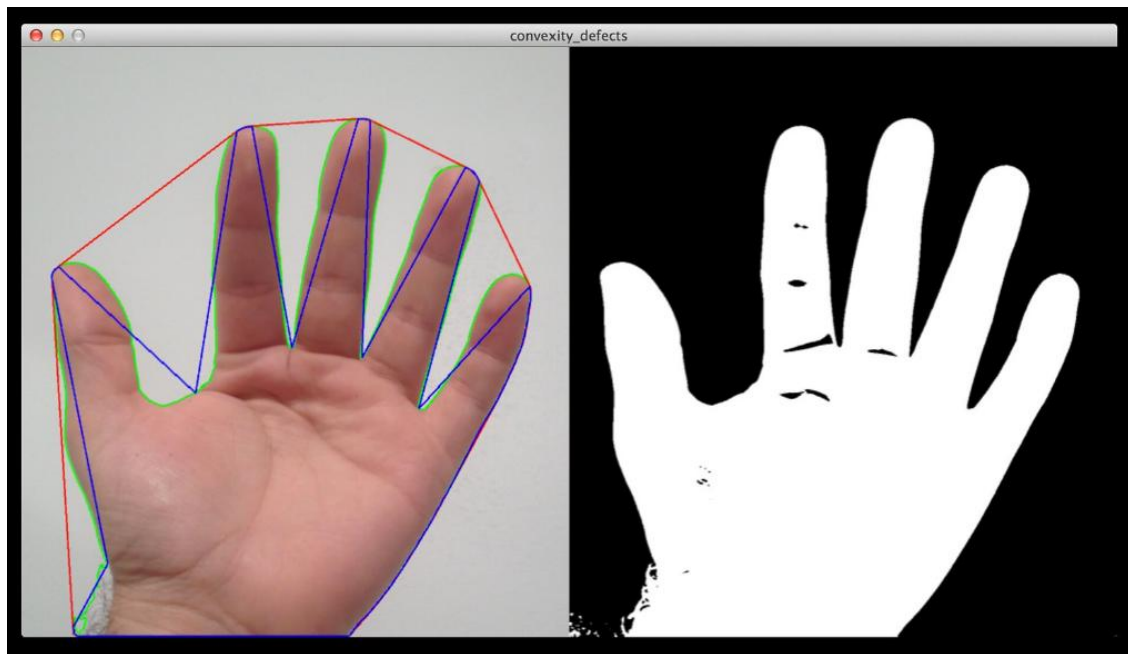


Image from: <https://www.flickr.com/photos/unavoidablegrain/9555538171>.

Thinning

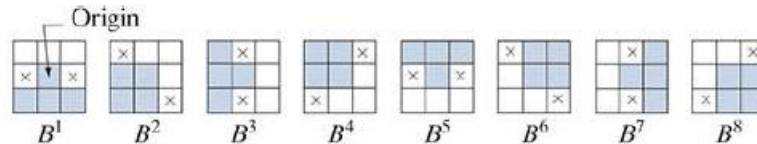
- ▶ Thinning of an object A in a binary image I can be defined as

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned}$$

- ▶ It can be extracted by repeating step

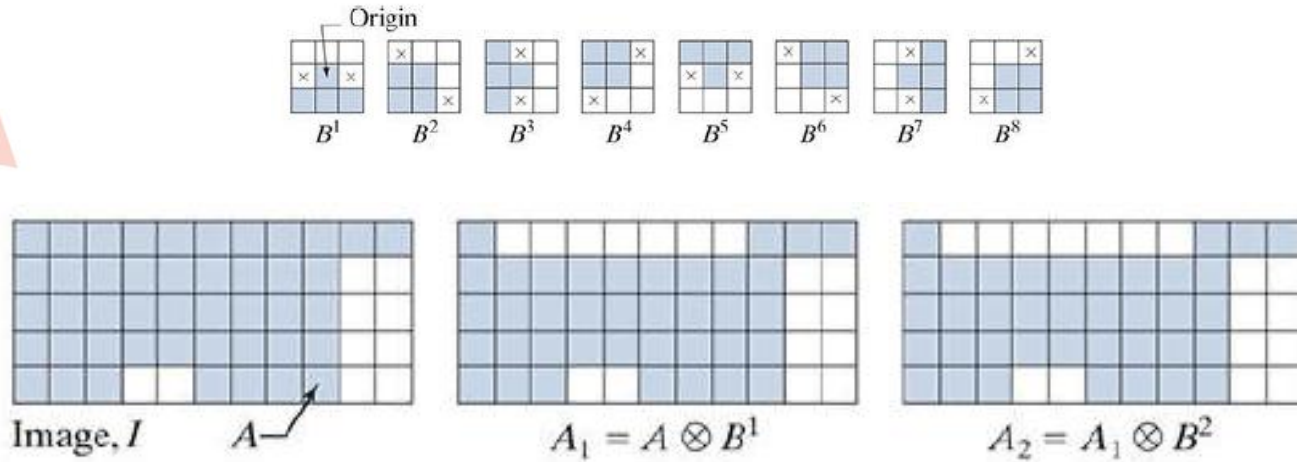
$$A \otimes \{B\} = (((((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$$

- ▶ with 8 SEs as follow

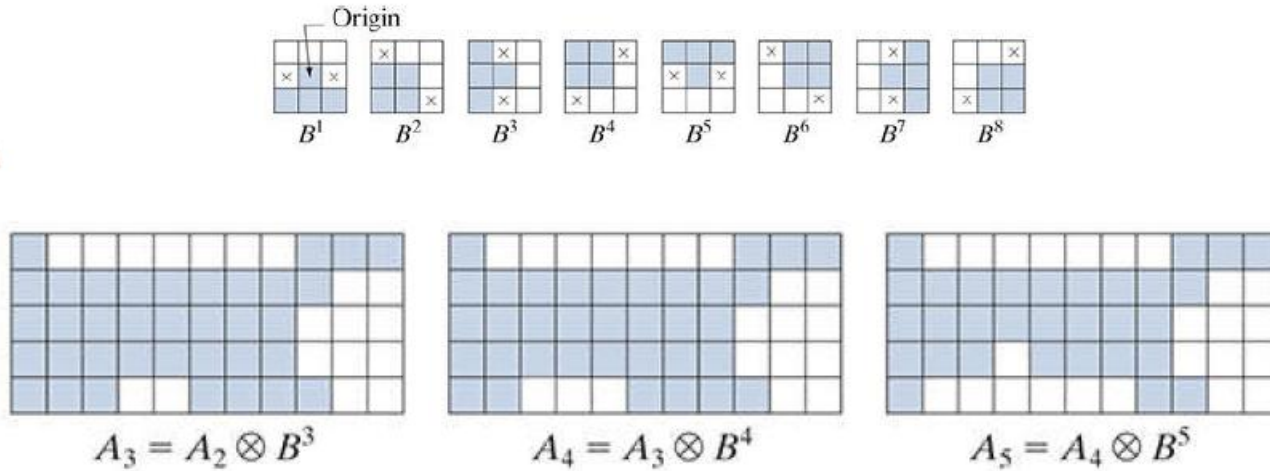


- ▶ Until every pass of SEs do not change the elements of the set

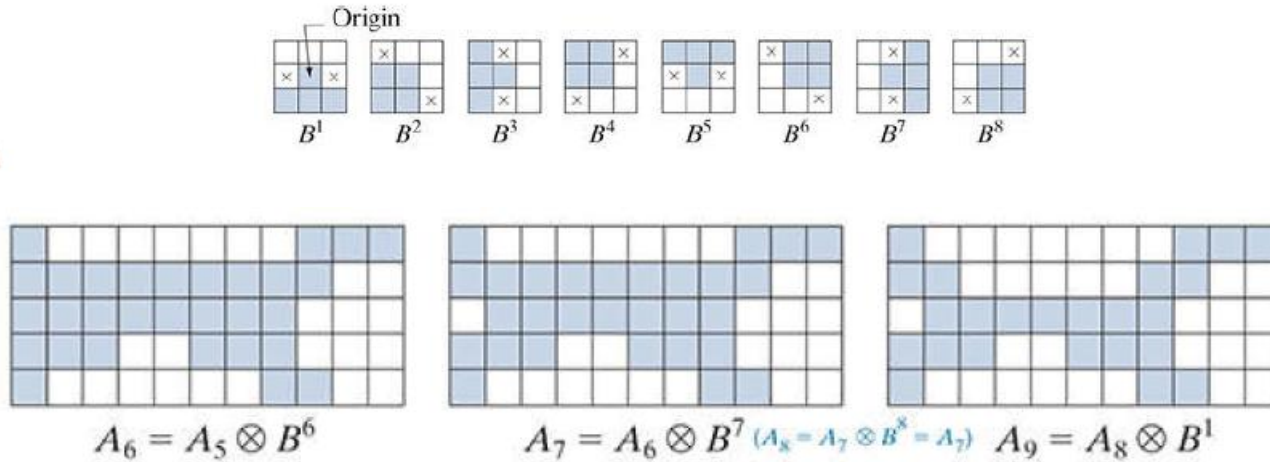
Thinning



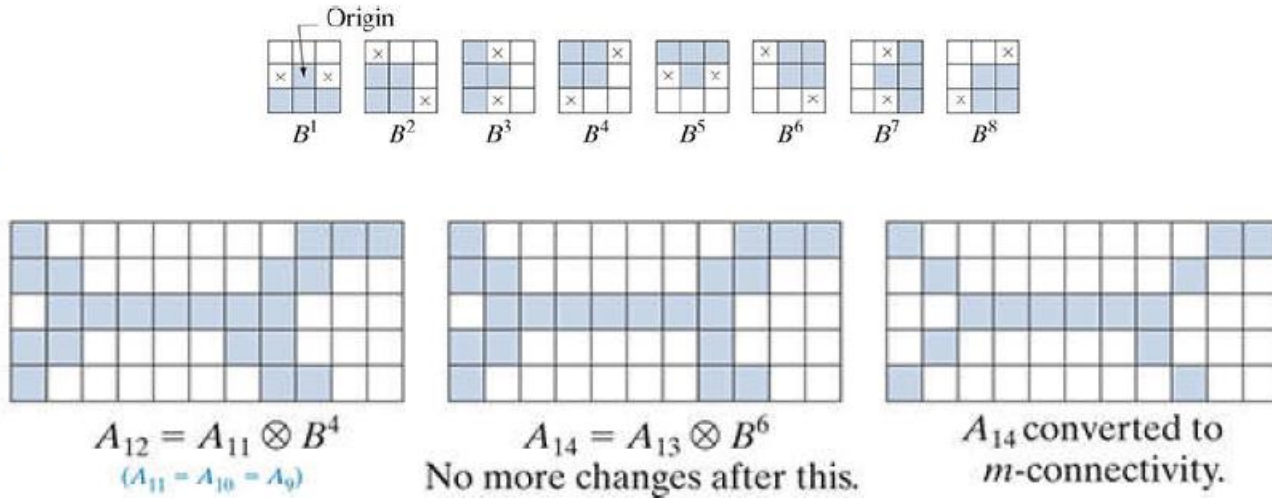
Thinning



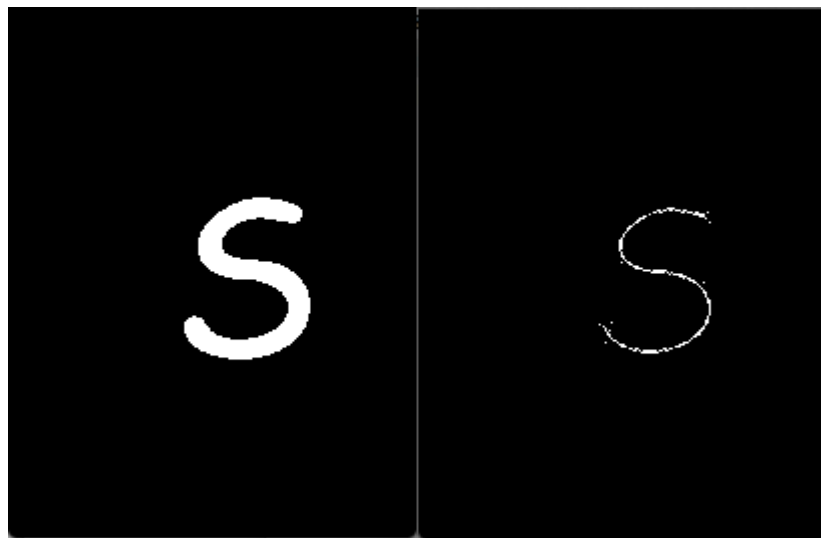
Thinning



Thinning



Thinning



Thickening

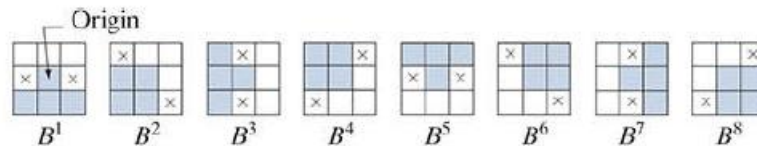
- ▶ Thickening of an object A in a binary image I can be defined as

$$A \odot B = A \cup (A * B)$$

- ▶ It can be extracted by repeating step

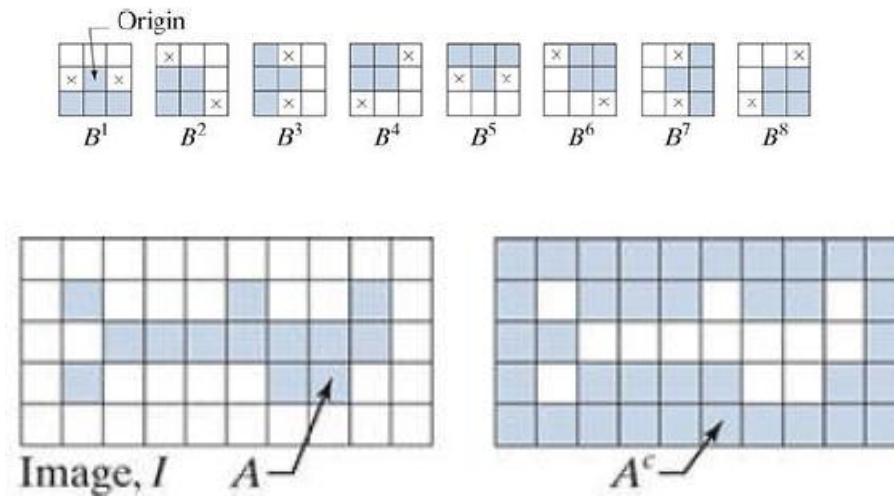
$$A \odot \{B\} = (((...((A \odot B^1) \odot B^2)...) \odot B^n)$$

- ▶ with 8 SEs as follow

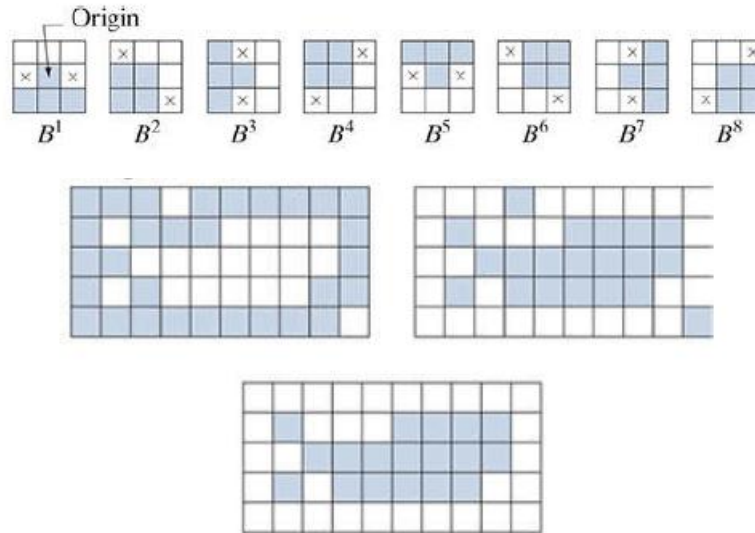


- ▶ Until every pass of SEs do not change the elements of the set

Thickening



Thickening



Thickening



Skeletons

- ▶ Skeleton of an object A in a binary image I can be defined as

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

- ▶ where $S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$

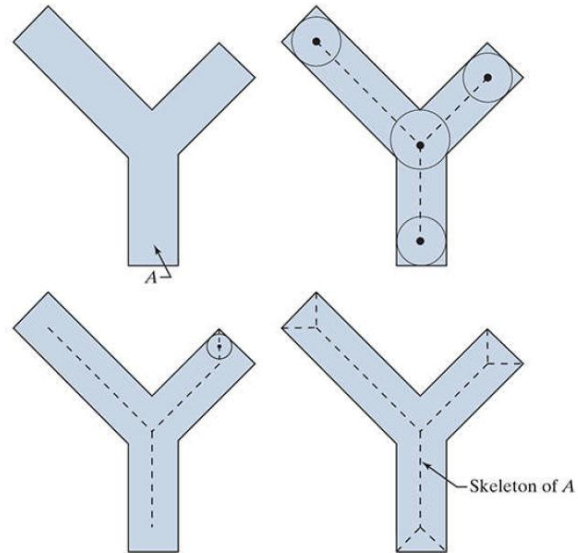
- ▶ with an SE  and $(A \ominus kB)$ indicates k successive erosions as

$$(A \ominus kB) = (((...(A \ominus B) \ominus B) \ominus ...) \ominus B)$$

- ▶ K is the last iterative step before A erodes to an empty set

$$K = \max\{k | (A \ominus kB) \neq \emptyset\}$$

Skeletons



Skeletons

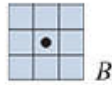
- ▶ From a skeleton $S(A)$, A can be reconstructed from

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

- ▶ where

$$(S_k(A) \oplus kB) = (((S_k(A) \oplus B) \oplus B) \oplus \dots) \oplus B)$$

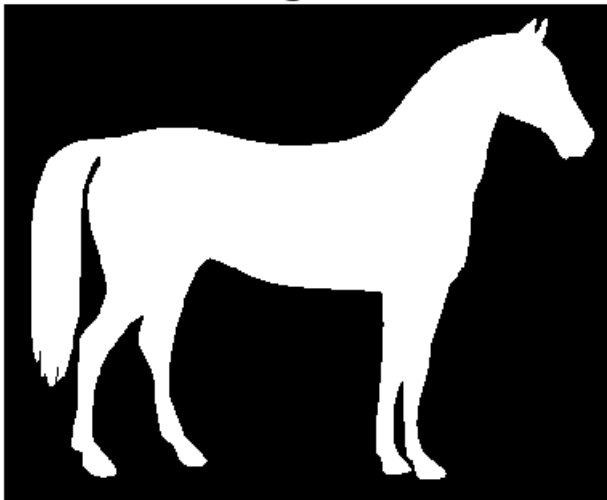
- ▶ with an SE dilations



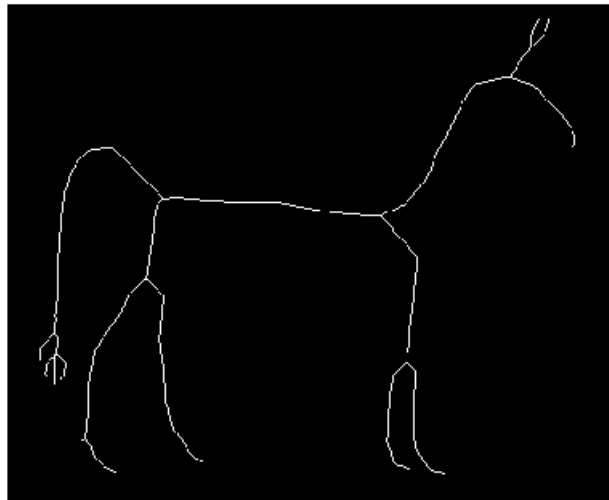
and $(S_k(A) \oplus kB)$ indicates k successive

Skeletons

original

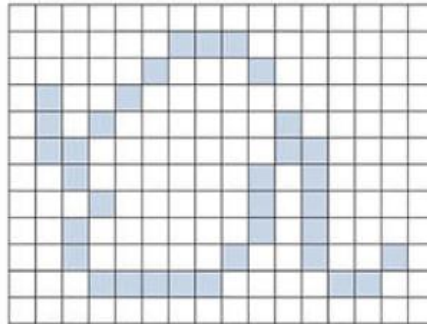


skeleton



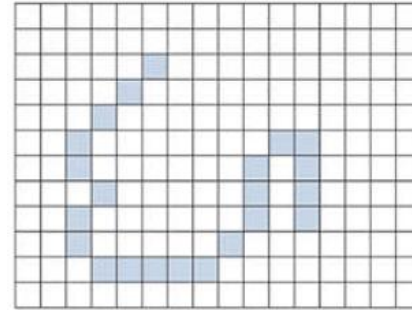
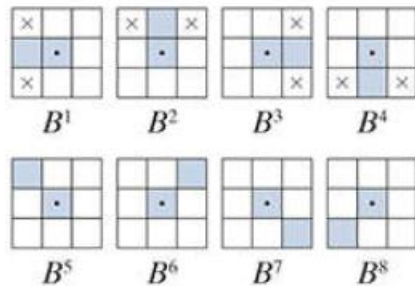
Pruning

- ▶ Pruning methods are essential complement to thinning and skeletonizing algorithms, because these procedures tend to leave spurs (“parasitic” components) that need to be “cleaned up” by postprocessing. We can identify spurs as limited group of pixels



Pruning

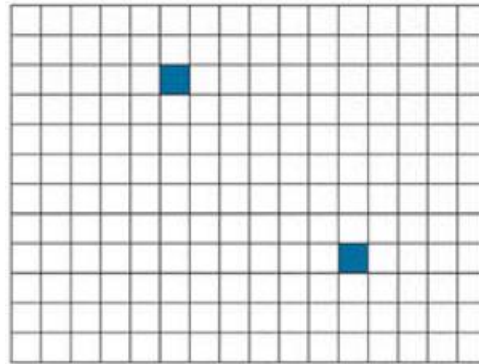
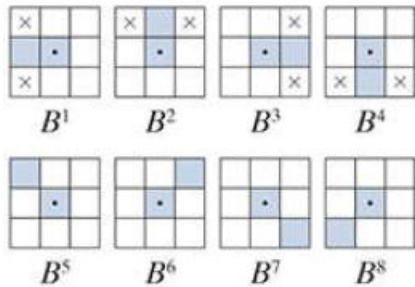
- First, identify the end points by thinning this sets of SEs according to the number of pixels which needs to be removed. In this case, we indicates that spurs are set of 3 or less pixels. Therefore, thin A 3 times $X_1 = A \otimes \{B\}$



Pruning

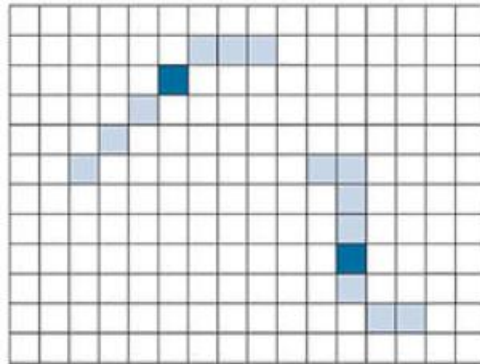
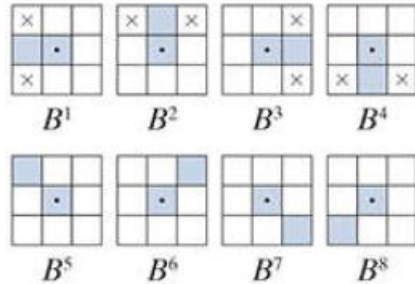
- Second, apply HMT with those SEs

$$X_2 = \bigcup_{k=1}^8 (X_1 \odot B^k)$$



Pruning

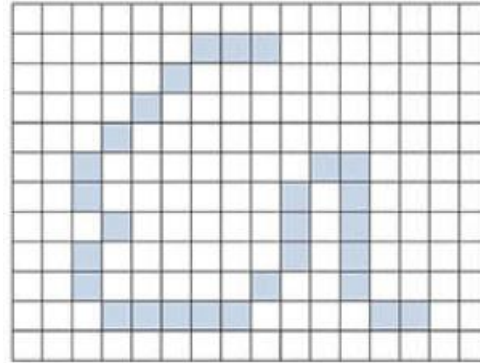
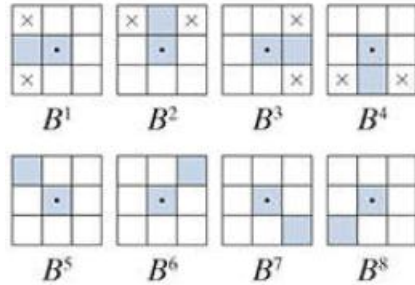
- ▶ Third, dilates the end points 3 times $X_3 = (X_2 \oplus H) \cap A$ which will guarantee that only end points will grow, not spurs



Pruning

- ▶ Last, union these end points back to the thinned set

$$X_4 = X_1 \cup X_3$$





Grayscale Erosion and Dilation

Grayscale Erosion and Dilation

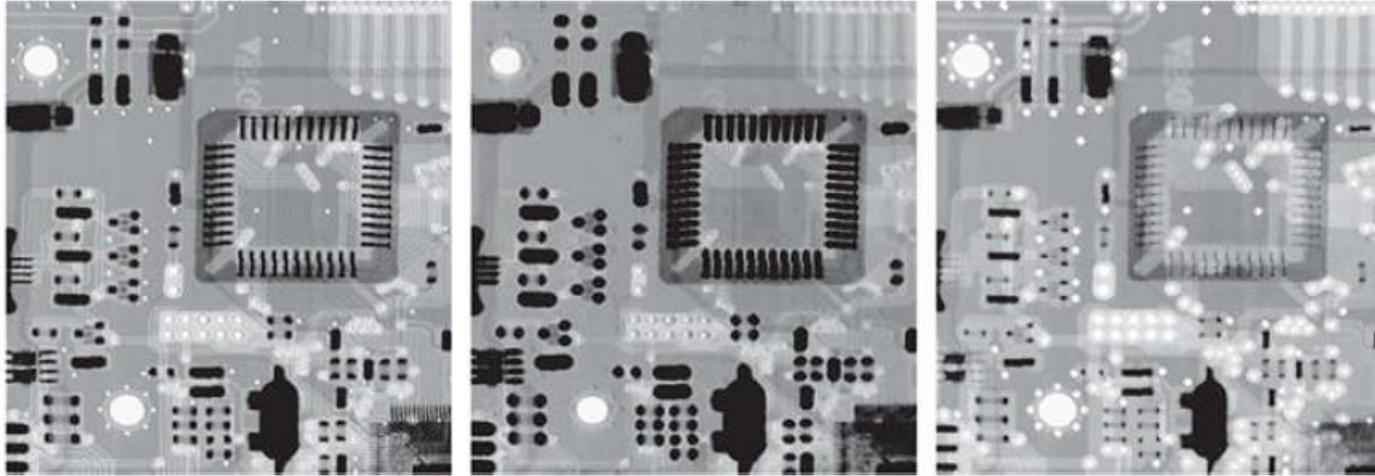
- ▶ Grayscale erosion of f can be obtained by follow the erosion of binary image. But instead, place the value of minimum intensity to the origin

$$[f \ominus b](x, y) = \min_{(s, t) \in b} \{f(x + s, y + t)\}$$

- ▶ On the other hand, grayscale dilation of f can be obtained by follow the dilation of binary image. But instead, place the value of maximum intensity to every pixels which are dilated

$$[f \oplus b](x, y) = \max_{(s, t) \in b} \{f(x - s, y - t)\}$$

Grayscale Erosion and Dilation





End of Week 8